AI Playbook and Toolkit for Public Health Departments

Data Quality Engineering for AI

OPS 200

Data quality engineering is the practical work of defining, measuring, improving, and monitoring data quality so that data are fit for use. In public health AI, data quality is not a background concern. It directly affects whether AI outputs are accurate, equitable, timely, and safe enough to support public health work.

Level	applied/foundational
Primary track	Operations and Data Quality Track
Appears in tracks	technical-architecture

Learning Objectives

- Identify major data quality dimensions relevant to public health AI.
- Explain the difference between general data quality and fitness for a specific use.
- Assess how missingness, timeliness, duplication, coding variation, and representativeness affect AI outputs.
- Define quality thresholds, review rules, and escalation criteria for a workflow.
- Produce a data quality review plan for an AI-supported use case.

Module Overview

Data quality engineering is the practical work of defining, measuring, improving, and monitoring data quality so that data are fit for use. In public health AI, data quality is not a background concern. It directly affects whether AI outputs are accurate, equitable, timely, and safe enough to support public health work.

Why This Topic Matters for Public Health AI

AI systems can create a false sense of precision when they produce polished outputs from weak data. A summary may sound authoritative even when key fields are missing. A risk score may look objective even when the data underrepresent communities with limited healthcare access. A classification model may appear efficient while encoding historical reporting patterns. Data quality engineering helps agencies identify these problems before they become operational decisions.

Public health data quality should always be evaluated in relation to the intended use. A dataset may be adequate for annual descriptive reporting but inadequate for real-time triage. A field may be useful for aggregate trends but too incomplete for individual-level decision support. Fitness for use is therefore a public health judgment as well as a technical assessment.

Data quality review also supports trust. Staff are more likely to use AI responsibly when they understand the limitations of the data behind the output. Leaders and communities are more likely to trust AI-supported workflows when limitations, uncertainty, and safeguards are documented.

Core Concepts

Completeness. Completeness refers to whether required data fields are present. Missing fields may affect case classification, risk prediction, subgroup analysis, and outreach planning.

Timeliness. Timeliness refers to whether data are available quickly enough for the intended use. Delayed data may be acceptable for retrospective analysis but unsafe for urgent response.

Validity. Validity refers to whether values follow expected formats, ranges, and rules. Invalid dates, codes, units, or categories can undermine AI performance.

Representativeness. Representativeness refers to whether data reflect the population, events, or settings relevant to the use case. Underrepresentation can produce biased or incomplete AI outputs.

Data drift. Data drift is a change in data patterns over time. Drift can occur when reporting partners, testing practices, disease patterns, coding rules, or community behaviors change.

Public Health Example

A health department may use AI to identify cases needing rapid follow-up. If demographic fields are missing more often for some populations, the model may be less reliable for those groups. If the data feed from certain clinics is delayed, the prioritization queue may reflect reporting timeliness rather than true need. A data quality review can reveal these issues before the tool is deployed.

A community health assessment team may use AI to summarize survey comments and administrative data. If survey responses are not representative of residents with limited English proficiency, unstable housing, disability, or limited internet access, the AI-generated themes may miss important community needs. Data quality engineering should include equity and representativeness review.

Technical Deep Dive

Data quality engineering begins by defining required fields and acceptable thresholds for the intended use. For an AI-supported case triage workflow, required fields may include condition, lab evidence, onset date, specimen date, jurisdiction, age, pregnancy status, hospitalization, and contact information. The agency should decide which missing fields stop automated support, which trigger a warning, and which can be handled through human review.

Quality measurement should include completeness, timeliness, validity, consistency, uniqueness, accuracy, and representativeness. Some dimensions can be measured automatically. Others require program knowledge, source review, or comparison to external benchmarks. Data quality should not be treated as a one-time predeployment activity. It should be monitored throughout the AI lifecycle.

Documentation is essential. A data quality plan should describe known limitations, review thresholds, monitoring frequency, responsible owners, escalation criteria, and decisions about acceptable use. When data quality is insufficient, the correct response may be to restrict the use case, add human review, improve the data pipeline, delay deployment, or avoid AI use for that workflow.

Risks, Failure Modes, and Guardrails

Confident output from weak data. AI may produce polished outputs from incomplete or biased data. Guardrails include uncertainty labeling, source review, and limits on use.

Fitness-for-use mismatch. Data suitable for one purpose may be unsafe for another. Guardrails include use-case-specific quality criteria and approval conditions.

Unmonitored drift. Data patterns may change after deployment. Guardrails include drift indicators, periodic review, and retraining or recalibration triggers.

Equity blind spots. Missingness and underrepresentation may affect some groups more than others. Guardrails include subgroup review, community context, and documentation of limitations.

Application to AI-Supported Workflows

Generative AI may summarize datasets or records, but the output must reflect missingness and uncertainty. Predictive analytics depend heavily on feature quality, label quality, and representative training data. NLP may be affected by language variation, abbreviations, and documentation practices. RAG depends on document quality, metadata, and currency. Agentic AI should be blocked from acting automatically when required data are missing or quality thresholds are not met.

Reflection Questions

Which data quality problems most often affect your program area?

Which missing fields would make an AI output unsafe or unreliable?

Which populations or settings may be underrepresented in the data?

How would staff detect data drift?

Who has authority to pause or limit AI use when data quality is insufficient?

Practical Exercise

Create a data quality review plan for one AI-supported workflow. Include the use case, required fields, quality dimensions, thresholds or review rules, known limitations, equity concerns, drift monitoring, documentation expectations, and escalation criteria.

Expected Artifact or Evidence

Data quality review plan

Data quality checklist

Thresholds or review criteria

Known limitations statement

Expected Assignment

- Data quality review plan
- Data quality checklist
- Thresholds or review criteria
- Known limitations statement

Knowledge Check

- 1. What does fitness for use mean?
- 2. Why can poor data quality be dangerous in AI-supported workflows?
- 3. What is data drift?
- 4. Which item belongs in a data quality review plan?
- 5. When should an AI use case be paused or limited?

References and Resources

- CDC surveillance and data quality resources
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- Agency data governance policies